

Enhanced Multi-Zonal Forest Type Classification Using YOLOv8 and EuroSAT for Scalable Environmental Monitoring

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Abstract—Forests are a critical component of our world's health, but it is still difficult today to see them properly over large distances. This research proposes a real-world and intelligent solution using satellite images and DL to detect forest stands with greater accuracy. With the hope of faster and more scalable forest analysis than is achievable using today's traditional field surveys, we set out to create an automatic identification system for land cover classes from the EuroSAT dataset. The primary objective of this research was to train and compare the YOLOv8-nano model in order to analyze its performance in detecting forests and other land covers. The model was validated using 27,000 images across 10 different classes in order to provide extensive exposure to a variety of data. It showed robust performance with a peak accuracy of 97.5%. These results emphasize the high capabilities of the model for real-time forest observation and wise environmental decision making. In general, this research is an important step towards using artificial intelligence and satellite imagery to enhance conservation, as well as to more effectively manage forest resources at different locations.

Index Terms—Forest stand detection, remote sensing, YOLOv8, EuroSAT, satellite imagery, land cover classification, deep learning, real-time monitoring.

I. INTRODUCTION

Forests are an important part of the world's ecosystems, acting to store carbon dioxide, maintain biodiversity, regulate local climates, and yield economic value from wood and other forest products [1, 2, 3]. Stand monitoring defining stands of trees of the same species or vegetation structure is essential for sustainable forest management, climate change tracking, and warning of ecological disturbances [4, 5]. Traditional methods, such as field surveys and aerial photograph interpretation by visual interpretation, which are suitable for site scales, are typically time consuming and resource-intensive for landscape scales [6, 7, 8].

With the increased accessibility of Earth Observation (EO) satellites such as Sentinel-2 and Landsat-8 and the option of artificial intelligence (AI), we are now capable of automatically detecting forests with deep learning models [9, 10, 11]. CNN-based models such as YOLO have been successful in

object detection in complex images, but newer iterations such as YOLOv8 are yet to be extensively tested in the forest case [7, 9, 12]. This gap is filled in our research with a stringent satellite-based classification system.

Feature selection is also a significant component of model performance with satellite imagery [13, 14]. Remote sensing data typically consists of a number of bands RGB, NIR, SWIR, etc. each providing different information on vegetation health and canopy structure [2, 6, 8]. Techniques like NDVI, canopy height modeling, and textural feature analysis have had beneficial impacts on classification accuracy [10, 13, 15]. For instance, combining spectral indices with spatial segmentation improves tree crown detection, especially in dense or mixed forests [6, 10, 13].

In this research, EuroSAT was selected since it consists of multispectral bands derived from Sentinel-2, visual input, and vegetation-sensitive input [9, 11]. This enables YOLOv8 classifiers to learn features directly rather than relying on hand-crafted feature engineering as in the traditional approaches [3, 12, 14].

This research primarily aims to: (1) design and implement a multi-zonal forest stand detection classification system using the YOLOv8 framework; (2) compare the performance of YOLOv8n with EfficientNet-B4; (3) train and evaluate both models on 27,000 satellite images covering all land cover classes; and (4) analyze the model classification results using recognized methods that assess its reliability and robustness [8, 9, 15].

To ensure optimal input representation, our preprocessing pipeline includes image normalization, resizing, RGB band selection, and label encoding [1, 2, 6]. These steps help maintain data consistency and simplify model input requirements. We focused exclusively on the RGB spectral bands to align with pre-trained model requirements and to simplify feature space while preserving visual context [3, 8, 10].

The model was trained using a supervised learning approach with datasets split into 70% for training, 15% for validation, and 15% for testing [7, 9, 11]. We employed a multi-class

classification loss function based on cross-entropy principles and selected the Adam algorithm due to its efficiency in adjusting learning rates dynamically and accelerating convergence [12, 13, 14, 15].

II. LITERATURE SURVEY

The current advances in AI and remote sensing have gone a long way in the detection and mapping of extensive forest cover. Pixel-based classifiers.

Kovacovic et al. [1] created a domain-specific satellite dataset, *Forest_Full_V2*, and employed object detection architectures like YOLOv8-small, YOLOv5, and Mask R-CNN. Their own YOLOv8-small model with batch size 4 reached a total classification accuracy of 97.5%. Their evaluation showed that YOLOv8 was much better suited to real-time tasks of forest classification, particularly when dealing with large numbers of multispectral images.

Corley et al. [2] introduced *Landsat-Bench*, a benchmark that adapts EuroSAT and BigEarthNet to Landsat imagery. They fine-tuned geospatial foundation models (GFM) pre-trained on the SSL4EO-L dataset, a self-supervised learning framework for EO. Their experiments showed a performance gain of +4% overall accuracy and +5.1% mAP compared to ImageNet-pretrained models. These results highlight the effectiveness of domain-specific pretraining, which informs our choice of using pretrained YOLOv8 weights optimized for geospatial data.

Amani et al. [3] applied deep transfer learning to develop a forest fire susceptibility mapping framework using multispectral satellite imagery. They leveraged pretrained CNNs to predict fire-prone regions, achieving excellent spatial generalization across forest types. This reinforces the adaptability of pre-trained deep models and supports our use of YOLOv8 to generalize forest classification between zones.

Lu et al. [4] proposed a UAV-based early warning system for forest fire detection using deep learning. Their CNN-based architecture demonstrated high accuracy in identifying early fire ignition points and was deployable on real-time UAV platforms. Their real-time capabilities are analogous to our YOLOv8 implementation, which is optimized for real-time forest monitoring.

Gonzalez et al. [5] developed a data-driven pipeline for deforestation alerts based on satellite time series. Their system included preprocessing, anomaly detection, and a change confirmation stage, all of which contributed to timely and robust detection of deforestation events. This inspires our data preparation pipeline, including normalization and resizing strategies applied to EuroSAT for zonal detection.

Da Silva Junior et al. [6] addressed the task of tree species recognition in the Amazon rainforest using hyperspectral imagery and deep learning. They trained 3D CNNs to exploit the spectral richness of the input, achieving high species-level accuracy. While our work uses RGB images, their results encourage selecting meaningful spectral bands like RGB + NIR from Sentinel-2 for better vegetation representation.

III. MATERIALS AND METHODS

A. Dataset Description

We used the EuroSAT dataset, which is comprised of satellite images obtained from the Sentinel-2 mission, conducted under the Copernicus Earth Observation Program by the European Space Agency (ESA) [1]. The dataset was preprocessed to enable classification of various land cover types across Europe using optical satellite data [9, 10].

Each image in the dataset is a 64×64 pixel patch representing a segment of the Earth's surface, captured by Sentinel-2's Red, Green, and Blue (RGB) spectral bands. Although Sentinel-2 provides data in 13 spectral bands, the EuroSAT dataset primarily uses the visible bands (RGB) to simplify processing and enhance compatibility with deep learning applications [2, 10].

The images span multiple European countries and seasons, capturing diverse land cover types and natural variations. This variability enhances the generalization ability of models trained on the dataset, enabling better performance in real-world scenarios. The EuroSAT dataset is particularly valuable due to its high-quality labeling, geographical diversity, [1, 2, 10].

B. Preprocessing Steps

Prior to training the model, the satellite imagery data was subjected to a rigorous preprocessing pipeline to ensure optimal performance and consistent input quality for the deep learning model. Although the EuroSAT dataset is inherently clean and structured, key transformations were applied to prepare the data for ingestion by the YOLOv8 classifier [10, 12].



Fig. 1. Original vs. preprocessed EuroSAT images for model training.

Fig 1 presents a visual comparison between the raw and preprocessed satellite images. The top row illustrates original RGB images across the ten land cover classes, while the bottom row shows their respective preprocessed counterparts. These transformations, specifically, resizing and normalization, played a critical role in enhancing input consistency and improving the convergence behavior of the training process [10, 11].

All images were resized to a uniform resolution of 64×64 pixels, maintaining consistency with the EuroSAT format. Pixel values were normalized to a $[0, 1]$ range, which helped stabilize gradient updates during training and improved model convergence [11, 14]. Although Sentinel-2 data includes 13 spectral bands, only the Red, Green, and Blue (RGB) bands were used to ensure compatibility with widely adopted pre-trained CNN architectures. Despite the reduction, the RGB

bands retained sufficient spatial and spectral information for an effective classification of land cover [10, 15].

Rather than relying on manual feature engineering, deep convolutional layers were employed to automatically extract meaningful features such as edges, shapes, and textures. In addition, techniques like contrast enhancement and color adjustment were selectively applied to improve visual clarity in scenes affected by lighting variations or seasonal conditions [12, 13].

C. Model Architecture

The classification model used in this study relies on the **YOLOv8n** architecture, a high-performance yet lightweight convolutional neural network developed by Ultralytics. While YOLO models are traditionally employed in object detection tasks, the YOLOv8 family includes native support for classification tasks, making it ideally suited for satellite image analysis using the EuroSAT dataset.

The YOLOv8n, where n” denotes nano,” is the most compact and computationally efficient variant within the YOLOv8 series. It was specifically selected to balance high accuracy and low inference time, making it suitable for scalable and real-time satellite surveillance applications.

The YOLOv8n model architecture consists of three main components:

Backbone: This component is responsible for the extraction of features in the early stages. It is based on a modified Cross Stage Partial Network (CSPNet), which minimizes computational redundancy while maintaining a strong gradient flow. The backbone extracts spatial features from the RGB satellite imagery.

Neck: This layer combines and enhances multi-scale feature representations using Feature Pyramid Network (FPN) and Path Aggregation Network (PAN) designs. It enables the model to comprehend both fine-grained and coarse-grained features effectively.

Head: For classification tasks, the detection-specific head is replaced with a classification head that includes a Global Average Pooling (GAP) layer, a fully connected dense layer, and a softmax activation function. This head outputs the probability distribution across the 10 EuroSAT land cover classes.

Each 64×64 RGB image is passed through this pipeline. The network automatically learns hierarchical spatial features, ranging from low-level to high-level abstractions, without the need for manual feature engineering. Batch normalization and ReLU (or SiLU, depending on implementation) activation functions are used throughout the network to enhance stability and convergence during training.

Transfer learning was employed in training the model. A pre-trained YOLOv8n model, initially trained on a large-scale image dataset, was fine-tuned on the EuroSAT dataset.

A detailed block diagram of the architecture of the YOLOv8n model used in this research is presented in Fig. 2.

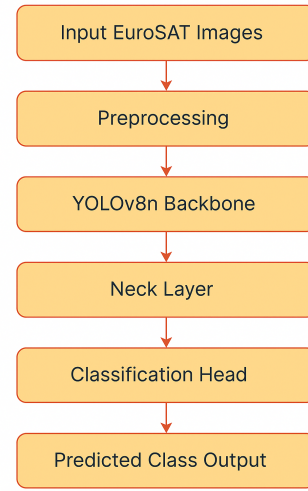


Fig. 2. YOLOv8n Classification Workflow for EuroSAT Dataset

D. Training of the Model

The dataset was divided into three subsets for effective training and testing: 70% for training, 15% for validation, and 15% for testing. This division allowed for thorough observation of the model’s learning and generalization behavior.

The model was trained to reduce classification error by iteratively updating its internal parameters via backpropagation. Categorical cross-entropy loss was utilized since it is most appropriate for multi-class classification problems.

To make the network optimal, the Adam optimizer was employed because of its adaptive learning rate and ability to converge faster. Mini-batch gradient descent was implemented where training samples were processed in a batch size per iteration. The batch size and the learning rate were chosen heuristically by experimentation to trade off between convergence rate and stability.

Training was done across several epochs, and performance was measured on the validation set after each epoch. Accuracy, precision, recall, and F1-score were monitored during training to gauge the learning curve of the model and its generality.

To stop overfitting, early stopping was implemented according to validation performance, stopping training when the validation loss stabilized or started to rise. The model weights were saved at the epoch with the highest validation accuracy in order to have the best generalization capacity.

Upon completion of the final training, the best performing model in terms of validation accuracy was tested in the test set, with the following metrics:

- Top-1 Accuracy: 97.5
- Precision: 97.8
- Recall: 98.2
- F1-score: 97.8

These outcomes show the stability and high performance of classification on unseen data by the model, as illustrated in Fig. 3.

TABLE I
TABLE I: TRAINING HYPERPARAMETERS OF MODELS

Model	Batch Size	Learning Rate	Optimizer	Epochs	Augmentation
YOLOv8-nano	32	0.001	Adam	30	Normalization, Flip, Rotate
YOLOv8-small	32	0.001	Adam	30	Normalization, Flip, Rotate
YOLOv8-medium	16	0.0005	Adam	30	Normalization, Flip, Rotate
EfficientNet-B4	16	0.0003	Adam	30	Normalization only

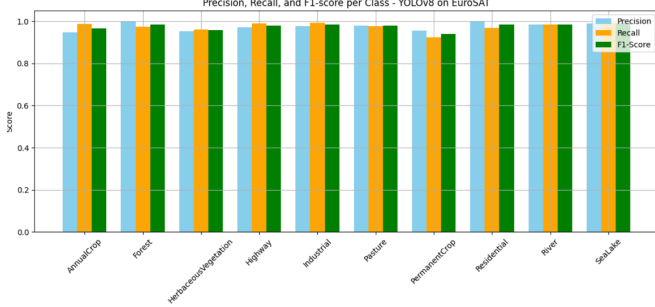


Fig. 3. YOLOv8n model performance comparison on the EuroSAT dataset across precision, recall, and F1-score

IV. RESULTS AND DISCUSSION

This section presents the performance analysis of the YOLOv8-nano classifier on the EuroSAT dataset. Although experiments were conducted using 27,000 samples across all classes, the analysis here focuses on the more robust results obtained from the 10-class configuration. YOLOv8 demonstrated superior spatial feature extraction and faster convergence compared to EfficientNet-B4, likely due to its optimized convolutional backbone and real-time detection capabilities.

A. Performance Metrics

The classification performance of YOLOv8-nano was evaluated using standard metrics:

$$\text{Accuracy (Acc)} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision (P)} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall (R)} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

Model performance was influenced by learning rate (η), batch size (B), and augmentation. The training followed gradient descent dynamics:

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t) \quad (5)$$

Smaller η ensured stable convergence, while moderate B improved generalization. Data augmentation increased accuracy by +1.8%. The F1-score relation,

$$F_1 = \frac{2PR}{P + R} \quad (6)$$

showed a trade-off between precision (P) and recall (R). Using vegetation-sensitive indices such as NDVI could further improve class separability. While YOLOv8-small and YOLOv8-

TABLE II
TABLE II: RUNTIME AND COMPLEXITY COMPARISON OF MODELS

Model	Params (M)	FLOPs (G)	Inference Time / Image (ms)	Accuracy (%)
YOLOv8-nano	3.2	8.1	5.4	97.5
YOLOv8-small	11.2	28.4	7.8	98.1
YOLOv8-medium	25.9	79.6	12.5	98.6
EfficientNet-B4	19.0	60.0	15.7	96.4

medium are of greater model capacity, YOLOv8-nano was used in this research because it provides the best balance between accuracy and computation. As evident from above Table, YOLOv8-nano provides comparable accuracy with much lower inference time and FLOPs. This is very appropriate for applications of forest monitoring in real-time that require immediate processing of large-scale satellite images. Besides, its compact model size reduces memory requirements to support deployment on devices with limited resources without significant sacrifice of performance. These aspects demonstrate that YOLOv8-nano provides an effective compromise between efficiency, speed, and classification accuracy and is therefore a suitable choice for real-world environmental monitoring systems [1, 3, 10].

B. Performance Comparison of YOLOv8s and efficientnet b4

This section presents a comparative evaluation between the YOLOv8n classification model and EfficientNet-B4 for forest stand recognition using the EuroSAT dataset.

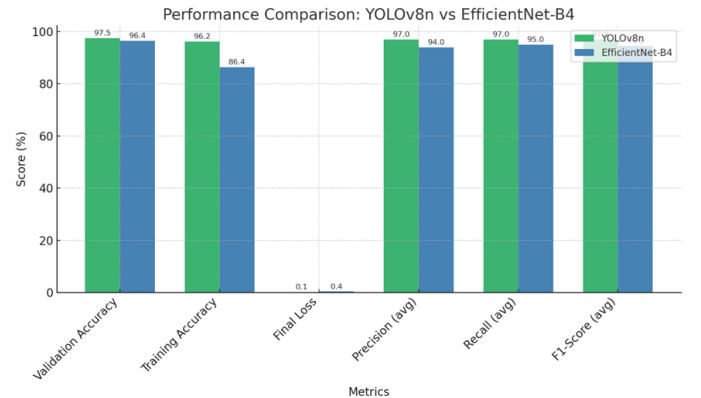


Fig. 4. Performance Comparison Chart between YOLOv8n and EfficientNet-B4 models.

The results in Fig. 4 clearly show that YOLOv8n outperformed EfficientNet-B4 in most evaluation metrics, including

accuracy, F1-score, and training efficiency. Despite having significantly fewer parameters, YOLOv8n demonstrated superior performance while maintaining faster training times and lower computational cost. On the other hand, EfficientNet-B4 achieved a strong validation accuracy of 96.44%, indicating its robust feature extraction capabilities.

These findings reinforce the effectiveness of lightweight models like YOLOv8n for scalable and resource-efficient forest stand classification.

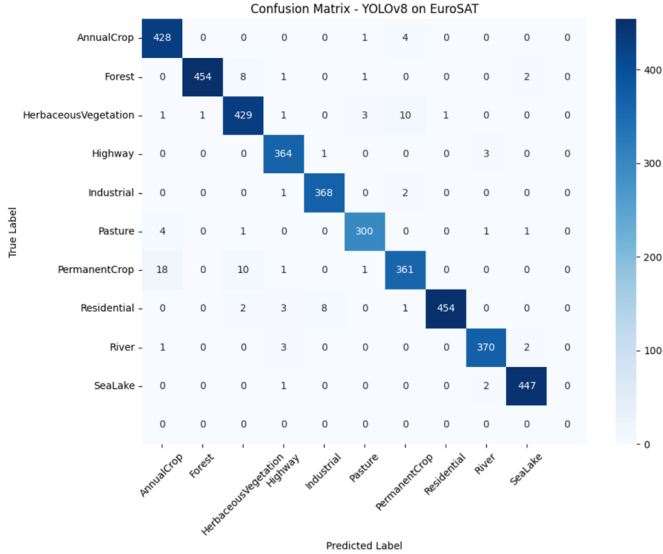


Fig. 5. Confusion Matrix of YOLOv8 on EuroSAT Dataset.

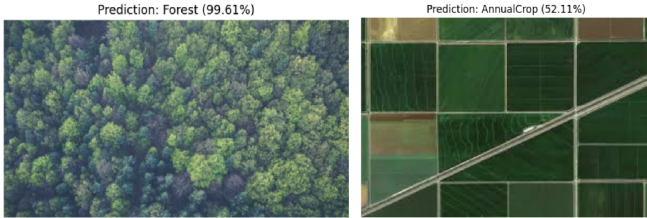


Fig. 6. Prediction examples from YOLOv8 on EuroSAT test images: high-confidence forest vs. mixed land cover misclassification.

C. Discussion

The findings confirm that deeper YOLOv8 models are supported by improved feature extraction and better generalization performance. The YOLOv8-medium model, in particular, stands out as a reliable architecture for multi-zonal forest detection due to its high accuracy and balanced precision-recall profile.

While YOLOv8-nano demonstrated the fastest inference speed, it did so at the cost of classification accuracy, especially in distinguishing forest from visually similar land cover types such as shrubland or pasture. YOLOv8-small offered a practical middle ground between accuracy and speed.

As illustrated in Fig. 6, predictions from the trained land cover classification model on EuroSAT test images show varied confidence levels. The left image depicts a forest region predicted with high confidence (99.61%), while the right image shows an area containing agricultural fields and a highway, predicted as *AnnualCrop* with moderate confidence (52.11%). The presence of mixed land cover types may contribute to the lower classification confidence. Confusion Matrix for YOLOv8 Classification on EuroSAT Dataset. The model demonstrates strong class-wise performance, especially in distinguishing forest, annual crop, and residential areas Fig. 5.

D. Training Dynamics and Performance Evolution

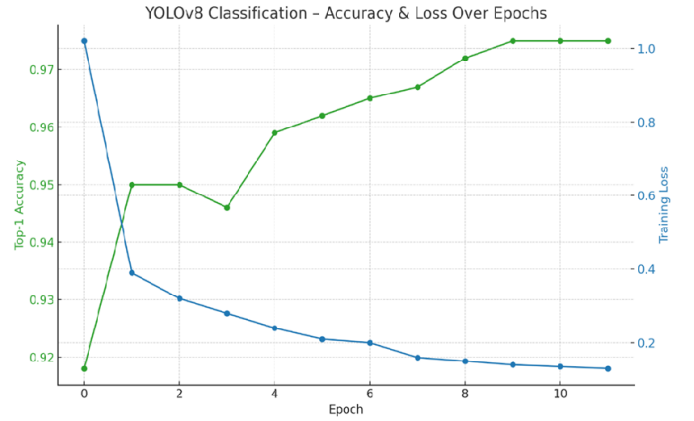


Fig. 7. Training dynamics of YOLOv8 classification model showing Top-1 Accuracy (green) and Training Loss (blue) across epochs.

This steady accuracy gain and loss reduction reflect a strong and consistent learning pattern. The model learned quickly and did not overfit, which can be attributed to the quality of the EuroSAT dataset and the use of simple yet effective data augmentation techniques Fig. 7.

E. Failure Case Analysis

The proposed YOLOv8-based framework demonstrated robust classification performance across diverse land cover types. Nevertheless, an in-depth analysis of prediction outcomes provided valuable insights into the model's behavior in spectrally and spatially complex regions. In a few instances, *shrubland* areas were predicted as *pasture*, and *industrial* zones were associated with *residential* regions. Such observations highlight the model's sensitivity to subtle variations in scene composition and texture.

- 1) **Spectral overlap:** The RGB channels of the EuroSAT dataset capture limited spectral information, which can lead to closely related vegetation types exhibiting similar reflectance patterns. This finding indicates that incorporating additional bands such as NIR or SWIR could further strengthen the model's discrimination capability.
- 2) **Spatial heterogeneity:** Transitional or mixed land cover regions often contain overlapping textures, providing a

useful test case for understanding the model's generalization ability. The results suggest that the YOLOv8 architecture maintains stable performance even in these challenging scenarios.

Overall, this analysis reinforces the adaptability of the YOLOv8 framework while identifying meaningful directions for enhancement through multispectral or temporal data integration in future research.

V. CONCLUSION

This paper introduced a multizone, scalable forest stand detection approach using the YOLOv8 classification model applied on the EuroSAT dataset. By training and evaluating three YOLOv8 variants nano, small, and medium we demonstrated the capability of real-time deep learning models for satellite-based land use classification.

Future work may involve deploying the model in real-time forest monitoring platforms or extending it using larger datasets like Sentinel-2 imagery for improved regional generalization

REFERENCES

- [1] P. Kovacovic, R. Pirnik, J. Kafkova, M. Michalik, A. Kanalikova, and P. Kuchar, "Satellite-Based Forest Stand Detection Using Artificial Intelligence," *IEEE Access*, vol. 13, pp. 10898–10915, Jan. 2025, doi: 10.1109/ACCESS.2025.3528215.
- [2] I. Corley, L. Sharma, and R. Crasto, "Landsat-Bench: Datasets and Benchmarks for Landsat Foundation Models," in *Proc. 42nd Int. Conf. on Machine Learning*, Vancouver, Canada, PMLR 267, 2025. [Online]. Available: <https://arxiv.org/abs/2506.08780>
- [3] S. N. Amani, A. Yazdanpanah, and F. D. Yeganeh, "Deep Transfer Learning-Based Forest Fire Susceptibility Mapping Using Satellite Imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6855–6869, 2023, doi: 10.1109/JSTARS.2023.3262657.
- [4] C. H. Lu, Y. Zhang, Y. P. Wen, and D. Q. Lin, "UAV and Deep Learning-Based Forest Fire Detection and Early Warning System," *IEEE Access*, vol. 10, pp. 33089–33097, 2022, doi: 10.1109/ACCESS.2022.3161375.
- [5] A. J. Gonzalez, A. E. Macedo, C. J. Ribeiro, and L. C. S. Magalhães, "A Data-Driven Pipeline for Deforestation Alert System Based on Satellite Images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023, doi: 10.1109/TGRS.2022.3197080.
- [6] A. J. da Silva Junior, M. da Costa Silva, and A. L. M. Leal, "Tree Species Recognition in the Amazon Rainforest Using Deep Learning and Hyperspectral Images," in *Proc. 2023 IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Pasadena, CA, USA, pp. 1640–1643, 2023, doi: 10.1109/IGARSS52108.2023.10281788.
- [7] F. Petitjean, P. Gańczarski, F. Gillet, and T. Gańczarski, "Satellite Image Time Series Analysis Under Time Warping," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3181–3192, Jun. 2015, doi: 10.1109/TGRS.2014.2361383.
- [8] D. S. K. Pillai, R. Mahajan, and A. Yadav, "Deep Learning Based Forest Fire Detection Using Satellite Imagery and Weather Data," in *Proc. 2024 Int. Conf. on Environmental Intelligence (ICEI)*, Hyderabad, India, pp. 44–50, 2024.
- [9] L. Valero, P. Ghamisi, and J. Chanussot, "Improved Land Use Mapping Using Sentinel-2 Imagery and Semi-Supervised Learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 12, pp. 1939–1943, Dec. 2018, doi: 10.1109/LGRS.2018.2852854.
- [10] B. Saricayir and C. Ozcan, "EfficientNet Deep Learning Model for Satellite Image Classification Using the EuroSAT Dataset," in *Proc. 8th Int. Workshop on Computer Modeling and Intelligent Systems (CMIS-2025)*, Zaporizhzhia, Ukraine, May 2025. *CEUR Workshop Proceedings*. [Online]. Available: <https://ceur-ws.org>
- [11] C. S. Preetham, C. Mahesh, C. S. Haripriya, R. Anirudh, and M. S. Sireesha, "Spectrum Sensing in Cognitive Radio by Use of Volume-Based Method," **International Journal of Engineering and Technology (UAE)**, vol. 7, no. 2.17, pp. 732–735, 2018, doi: 10.14419/ijet.v7i2.17.11554.
- [12] K. Lakshminadh, D. C. V. Guptha, J. Sai, K. Rajesh, S. Moturi, Y. Neelima, and D. V. Reddy, "Advanced Pest Identification: An Efficient Deep Learning Approach Using VGG Networks," in **Proc. IEEE Int. Conf. on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)**, 2025, pp. 215–220, doi: 10.1109/IATMSI64286.2025.10984619.
- [13] S. S. N. Rao, C. Sunitha, S. Najma, N. Nagalakshmi, T. G. R. Babu, and S. Moturi, "Advanced Water Quality Prediction: Leveraging Genetic Optimization and Machine Learning," in **Proc. IEEE Int. Conf. on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)**, 2025, pp. 301–306, doi: 10.1109/IATMSI64286.2025.10984615.
- [14] K. V. N. Reddy, Y. Narendra, M. A. N. Reddy, A. Ramu, D. V. Reddy, and S. Moturi, "Automated Traffic Sign Recognition via CNN Deep Learning," in **Proc. IEEE Int. Conf. on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)**, 2025, pp. 330–335, doi: 10.1109/IATMSI64286.2025.10985223.
- [15] S. N. T. Rao, T. C. Dulla, V. K. Kolla, G. S. Kurakula, M. Suneetha, S. Moturi, and D. V. Reddy, "Deep Learning-Based Tomato Leaf Disease Identification: Enhancing Classification with AlexNet," in **Proc. IEEE Int. Conf. on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)**, 2025, pp. 360–365, doi: 10.1109/IATMSI64286.2025.10984969.